

Drawdown Distribution around Wells Partially Penetrating Thick Leaky Artesian Aquifers

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Abstract. The differential equation governing nonsteady flow to a well in a thick leaky system has been solved by using transform techniques. Routines to tabulate the most general solution in type curve form have been developed and are available. The limiting curve of the tabulated function is the Hantush partial penetration type curve. With knowledge of the geologic framework the technique can be applied to certain field problems.

Leaky aquifers have been recognized since before the turn of the century [Jacob, 1946]. However, it has been only in the last 25 years that theoreticians have been able to define and solve special forms of this problem. In 1946, C. E. Jacob solved a steady state form of the differential equation for the leaky aquifer completely penetrated by a well. Leaky aquifer theory was actively pursued by Hantush [e.g., 1956, 1957, 1960; Hantush and Jacob, 1954, 1955], Neuman and Witherspoon [1969], Herrera and Figueroa [1969], and Herrera [1970]. The solutions by Hantush and Jacob were for completely or partially penetrating wells, and the leakage term was included in the differential equation. Including the leakage or source term in the differential equations assumes that the upper and lower boundaries are impermeable and that the leakage is generated within the aquifer at a rate directly proportional to the drawdown at any point. The use of this equation for partially penetrating wells involves error, the magnitude of which increases as the ratio of penetration to the total thickness decreases. The work by Neuman and Witherspoon and by Herrera and Figueroa was for completely penetrated leaky systems. Hantush [1967] solved the equation governing flow in a partially penetrated leaky aquifer by using the leakage term as a boundary condition. Solving the same differential equation by using slightly different bound-

ary conditions allows the solution to be applied over a wider range of conditions.

In an aquifer that is partially penetrated by a well, if the effects of pumpage are negligible at the lower confining layer of the aquifer, it is in effect infinitely thick. A leaky aquifer is treated as one receiving 'linear' leakage, or leakage that is directly proportional to the drawdown at the interface between the aquifer and the overlying leaky semipervious layer.

The system under consideration is that of a 'thick' aquifer overlain by a thin semipervious leaky layer, which in turn is overlain by a pervious saturated bed. The head above the semipervious leaky layer is considered constant, and storage in the semipervious layer is negligible. Leakage through the semipervious layer is vertical and proportional to the drawdown of the piezometric surface at the leaky interface. Flow to the well is considered axisymmetric, and discharge from the well constant. The aquifer is assumed to be homogeneous, isotropic, infinite in areal extent, and thick, and the drawdown distribution is initially 0 throughout the aquifer. The properties of the porous medium, namely, hydraulic conductivity and specific storage, remain constant in time and space.

FLOW SYSTEMS

Four flow systems (Figure 1) are treated for both unsteady and steady state flow:

1. Wholly perforated wells of penetration l' are treated as the purely axisymmetric flow of

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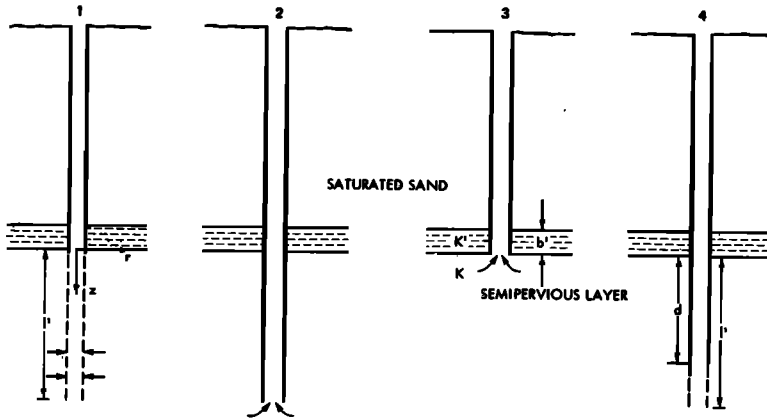


Fig. 1. Diagrammatic representation of the four flow systems. (1) Wholly perforated wells of penetration l' . (2) Wells of penetration l' and zero perforation. (3) Wells of zero penetration. (4) Wells of penetration l' and perforation $l' - d$.

water discharging into a cylinder of length l' .

2. Wells of penetration l' and zero perforation are considered as the purely radial flow of water into a spherical cavity of the same radius as the well.

3. Wells of zero penetration are treated as the purely radial flow of water to a hemispherical cavity that has the same radius as the well.

4. Wells of penetration l' and perforation ($l' - d$) are treated by superimposing a recharging well of penetration d on a discharging well of penetration l' , both wholly perforated.

Flow System 1: Wells of Finite Depth

Unsteady state. This system assumes a penetration l' equal to perforation. The system is axisymmetric, the origin being taken at the intersection of the axis of the well and the bottom of the semipervious layer. The boundary value problem in (r, z, t) is

$$\frac{\partial^2 s}{\partial r^2} + \frac{1}{r} \frac{\partial s}{\partial r} + \frac{\partial^2 s}{\partial z^2} = \frac{1}{\nu} \frac{\partial s}{\partial t} \quad z > 0 \quad (1)$$

$$\frac{\partial s}{\partial z}(r, 0, t) = \frac{s(r, 0, t)}{a} \quad (2)$$

$$\frac{\partial s}{\partial z}(r, \infty, t) = 0 \quad (3)$$

$$s(r, z, 0) = 0 \quad (4)$$

$$s(\infty, z, t) = 0 \quad (5)$$

$$\lim_{r \rightarrow 0} r \frac{\partial s}{\partial r} = -\frac{Q}{2\pi K l'} \quad 0 < z < l' \quad (6)$$

$$\lim_{r \rightarrow 0} r \frac{\partial s}{\partial r} = 0 \quad z > l'$$

in which

- 1/a, leakage factor, equal to $K'/b'K$, $1/L$;
- b' , thickness of the semipervious layer, L ;
- K , hydraulic conductivity of the aquifer, L/T ;
- K' , hydraulic conductivity of the semipervious layer, L/T ;
- l' , length of penetration of the well, L ;
- Q , discharge, L^3/T ;
- r , radial distance from the axis of the pumping well to any point in space, L ;
- s , drawdown at any point in the aquifer at any time t since pumping began, L ;
- S , storage coefficient, dimensionless;
- S_s , specific storage, $1/L$;
- t , time, T ;
- T , transmissivity, L^2/T ;
- z , vertical coordinate, L ;
- $\nu = K/S_s = T/S$, L^2/T .

By means of the Laplace transform with respect to t and (4), the Fourier cosine transform with respect to z and (2) and (3), and the Hankel transform of zero order with respect to r and (5) and (6), equation 1 will ultimately yield [e.g., Carslaw and Jaeger, 1959; Hantush, 1960].

$$\begin{aligned} & \bar{V}(\bar{\alpha}, \omega, P) \\ &= \frac{Q}{2\pi K l' p} \left(\frac{z}{\pi} \right)^{1/2} \frac{\sin \omega l'}{\omega[(p/\nu) + \omega^2 + \bar{\alpha}^2]} \\ & - \frac{1}{a} \left(\frac{z}{\pi} \right)^{1/2} \frac{\bar{V}(r, 0, P)}{(p/\nu) + \omega^2 + \bar{\alpha}^2} \end{aligned} \quad (7)$$

where p is time t transformed with respect to the Laplace transform, $\bar{V}$ is the zero order Hankel transform of $\bar{s}$, ω is z transformed with respect to the Fourier cosine transform, and $\bar{\alpha}$ is r transformed with respect to the zero order Hankel transform. By means of the Fourier cosine inversion formula, (7) becomes

$$\bar{V} = \frac{A}{p} \frac{2}{\pi} \int_0^\infty \frac{\sin(\omega l') \cos(\omega z) d\omega}{\omega[(p/\nu) + \omega^2 + \bar{\alpha}^2]} - \frac{1}{a} \frac{2}{\pi} \bar{V}(\bar{\alpha}, 0, p) \int_0^\infty \frac{\cos \omega z d\omega}{(p/\nu) + \omega^2 + \bar{\alpha}^2} \quad (8)$$

In the process of simplification the following relations have been used:

$$\int_0^\infty \frac{\sin(xy) dx}{x(x^2 + a^2)} = (\pi)^{1/2} \frac{1 - e^{-ay}}{a^2} \quad y > 0$$

and

$$\int_0^\infty \frac{\cos(xy) dx}{(x^2 + a^2)} = (\pi)^{1/2} \frac{e^{-ay}}{a}$$

By means of the inverse Hankel transform, (8) becomes

$$\bar{s} = \frac{A}{2} \int_0^\infty \frac{\{2 - \exp[-(l' + z)((p/\nu) + \bar{\alpha}^2)^{1/2}] - \exp[-(l' - z)((p/\nu) + \bar{\alpha}^2)^{1/2}]\} \bar{\alpha} J_0(\bar{\alpha} r) d\bar{\alpha}}{p[(p/\nu) + \bar{\alpha}^2]} - A \int_0^\infty \frac{(1/a) \{ \exp[-z((p/\nu) + \bar{\alpha}^2)^{1/2}] - \exp[-(l' + z)((p/\nu) + \bar{\alpha}^2)^{1/2}] \} \bar{\alpha} J_0(\bar{\alpha} r) d\bar{\alpha}}{p[(p/\nu) + \bar{\alpha}^2] \{ (1/a) + [(p/\nu) + \bar{\alpha}^2]^{1/2} \}} \quad (9)$$

where J_0 is a zero order Bessel function of the first kind.

By rearranging and using the convolution and translation theorems [e.g., Churchill, 1958], equation 9 is reduced to

$$s = \frac{Q}{8\pi K l'} \left\{ 2 \int_u^\infty \frac{e^{-y}}{y} e^{z/a} e^{r^2/4a^2 y} \cdot \operatorname{erfc} \left[\frac{r}{2a(y)^{1/2}} + \frac{z(y)^{1/2}}{r} \right] dy - 2 \int_u^\infty \frac{e^{-y}}{y} e^{(l'+z)/a} e^{r^2/4a^2 y} \cdot \operatorname{erfc} \left[\frac{r}{2a(y)^{1/2}} + \frac{(l' + z)(y)^{1/2}}{r} \right] dy \right.$$

$$\left. + 2M\left(u, \frac{z}{r}\right) + M\left(u, \frac{l' - z}{r}\right) - M\left(u, \frac{l' + z}{r}\right) \right\} \quad (10)$$

where $u = r^2/4\nu\tau$ and $M(u, c)$ is a function available in tabular form [Hantush, 1960] and is defined by

$$M(u, c) = \int_u^\infty \frac{e^{-y}}{y} \operatorname{erf}[c(y)^{1/2}] dy$$

where c is an argument in terms of l' , z , and r and erf and erfc are the error and complementary error functions, respectively. Equation 10 is the unsteady state final solution when $l' > z$. When $l' < z$, a process similar to that followed previously will yield the same result, namely, (10).

Steady state. By means of the final value theorem from the Laplace transforms [e.g., Churchill, 1958] the steady state solution to system 1 is

$$s = \frac{Q}{4\pi K l'} [c + 2I(\alpha, \beta) - 2I(\gamma, \beta)] \quad (11)$$

where

$$I(\alpha, \beta) = e^\alpha \int_\alpha^\infty \frac{e^{-\nu}}{(\beta^2 + \nu^2)^{1/2}} d\nu$$

$$I(\gamma, \beta) = e^\gamma \int_\gamma^\infty \frac{e^{-\nu}}{(\beta^2 + \nu^2)^{1/2}} d\nu$$

where

$$c = \left[\sinh^{-1} \left(\frac{l' - z}{r} \right) - \sinh^{-1} \left(\frac{l' + z}{r} \right) + 2 \sinh^{-1} \left(\frac{z}{r} \right) \right] \\ \alpha = z/a \quad \beta = r/a \quad \gamma = (z + l')/a$$

In simplifying, the following relation has been used:

$$\int_0^\infty x^{-3/2}(1 - e^{-ax}) J_\nu(xy)(xy)^{1/2} dx = (y)^{1/2} \sinh^{-1}\left(\frac{a}{y}\right) \quad z > 0 \quad \text{Re } a > 0$$

Flow System 2: Flow to a Spherical Cavity at a Depth of l'

Unsteady state. The unsteady state solution of flow to a well perforated throughout its depth of penetration, discussed above, was the result of averaging the flow to a spherical cavity over the length l' . Differentiating with respect to l' will therefore give the solution for flow system 2 if Q is replaced with $q l'$, where q is the discharge per unit area, L^2/T .

Differentiating (9) or the corresponding equation for $z > l'$ with respect to l' yields

$$\begin{aligned} \bar{s} = \frac{B}{2} \left[\int_0^\infty \frac{\exp \{-(l' + z)[(p/\nu) + \bar{\alpha}^2]^{1/2}\} \bar{\alpha} J_0(\bar{\alpha} r) d\bar{\alpha}}{p[(p/\nu) + \bar{\alpha}^2]^{1/2}} \right. \\ \left. + \int_0^\infty \frac{\exp \{-(l' - z)[(p/\nu) + \bar{\alpha}^2]^{1/2}\} \bar{\alpha} J_0(\bar{\alpha} r) d\bar{\alpha}}{p[(p/\nu) + \bar{\alpha}^2]^{1/2}} \right. \\ \left. - 2 \int_0^\infty \frac{(1/a) \exp \{-(l' + z)[(p/\nu) + \bar{\alpha}^2]^{1/2}\} \bar{\alpha} J_0(\bar{\alpha} r) d\bar{\alpha}}{p[(p/\nu) + \bar{\alpha}^2]^{1/2} \{ (1/a) + [(p/\nu) + \bar{\alpha}^2]^{1/2} \}} \right] \quad (12) \end{aligned}$$

where $B = q/2\pi K$, q being the discharge of the cavity.

By rearranging, using the convolution and translation theorems, and simplifying, (12) is reduced to

$$\begin{aligned} s = \frac{q}{4\pi K} \left\{ \frac{1}{R} \operatorname{erfc}(U) + \frac{1}{R'} \operatorname{erfc}(U') \right. \\ \left. - \frac{1}{a} \int_u^\infty \frac{e^{-y}}{y} e^{(l'+z)/a} e^{r^2/4a^2y} \right. \\ \left. \cdot \operatorname{erfc} \left[\frac{r}{2a(y)^{1/2}} + \frac{(l' + z)(y)^{1/2}}{r} \right] dy \right\} \quad (13) \end{aligned}$$

in which $R = [r^2 + (l' + z)^2]^{1/2}$, $R' = [r^2 + (l' - z)^2]^{1/2}$, $U = R/(4\nu\tau)^{1/2}$, and $U' = R'/(4\nu\tau)^{1/2}$. Equation (13) is the final unsteady state solution.

Steady state. The steady state solution is obtained from (13) at $t \rightarrow \infty$. As $t \rightarrow \infty$, $\operatorname{erf}(\infty) = 1$, and the third term becomes

$$\begin{aligned} s = \frac{B}{2} \int_0^\infty \left[e^{-(l'+z)\bar{\alpha}} + e^{-(l'-z)\bar{\alpha}} \right. \\ \left. - \frac{2(1/a)e^{-(l'+z)\bar{\alpha}}}{(1/a) + \bar{\alpha}} \right] J_0(\bar{\alpha} r) d\bar{\alpha} \quad (14) \end{aligned}$$

It can be shown from the solutions of completely perforated wells that the integral of (14) is equal to $I(\gamma, \beta)$. Consequently, the steady state solution becomes

$$s = \frac{q}{4\pi K} \left[\frac{1}{R} + \frac{1}{R'} - \frac{2}{a} I(\gamma, \beta) \right] \quad (15)$$

Flow System 3: Flow to a Hemispherical Cavity of Zero Penetration

The solutions to this system are obtained from those of the spherical cavity with $l' = 0$.

Unsteady state. The unsteady state solution to this system, obtained from (13) with $l' = 0$, yields

$$\begin{aligned} s = \frac{q}{2\pi K} \left\{ \frac{1}{R_0} \operatorname{erfc} \left[\frac{R_0}{(4\nu\tau)^{1/2}} \right] \right. \\ \left. - \frac{1}{a} \int_u^\infty \frac{e^{-y}}{y} e^{z^2/a} e^{r^2/4a^2y} \right. \\ \left. \cdot \operatorname{erfc} \left[\frac{r}{2a(y)^{1/2}} + \frac{z(y)^{1/2}}{r} \right] dy \right\} \quad (16) \end{aligned}$$

where $R_0 = r^2 + z^2$.

Steady state. Equation (15) with $l' = 0$ gives

$$s = \frac{q}{2\pi K} \left[\frac{1}{R_0} - \frac{1}{a} I(\alpha, \beta) \right] \quad (17)$$

Flow System 4: Well of Finite Depth and Perforation ($l' - d$)

Unsteady state. The unsteady state solution to this system is

$$s = [s(l') - s(d)] \quad (18)$$

where $s(l')$ is given by equation 10 (the draw-down equation for complete perforation over l') and $s(d)$ by the same equation with d replacing l' .

Steady state. The steady state solution to this system is

$$s = [M(l') - M(d)] \quad (19)$$

where $M(l')$ is given by (11) and $M(d)$ is given by (11) with l' replaced by d .

DISCUSSION

The preceding analysis solves four general flow systems, each corresponding to a different method of well completion. Because a well perforated throughout the zone of completion is most often encountered in the field, special attention has been given to evaluating (10) in a usable form (Figure 2).

The leaky partial penetration function $W(u, z/r, r/l', z/a)$ can be tabulated extensively to produce a large number of type curves. To give the investigator a high degree of freedom (Stallman, personal communication, 1971), a computer program [Hartman and Halepaska, 1971] is available to generate the necessary

sets of type curves for a specific set of parameters.

With reference to Figure 2 and on the assumption that a field well is compatible with the functions $z/r = 0.1$ and $r/l' = 0.5$, a match with one of the curves automatically defines the function z/a . From the match points, values of K and S are defined. When the thickness of the leaky layer from well logs, the value of K from the type curve match, and the value of a from z/a are known, the conductivity of the leaky layer can be calculated from the definition of a , i.e., $a = b'K/K'$.

As b' becomes large, the assumption of no storage in the leaky layer becomes increasingly invalid. The definition of a cannot be used under this circumstance to determine the conductivity of the confining bed. The limitations of assuming no storage in the leaky layer and zero draw-down in the layer above the leaky layer are well defined by Neuman and Witherspoon [1969] for a completely penetrated system. The limitations defined by Neuman and Witherspoon can be relaxed somewhat for the flow systems presented as the thicknesses of the system above the leaky layer and the produced aquifer become large with respect to the leaky layer.

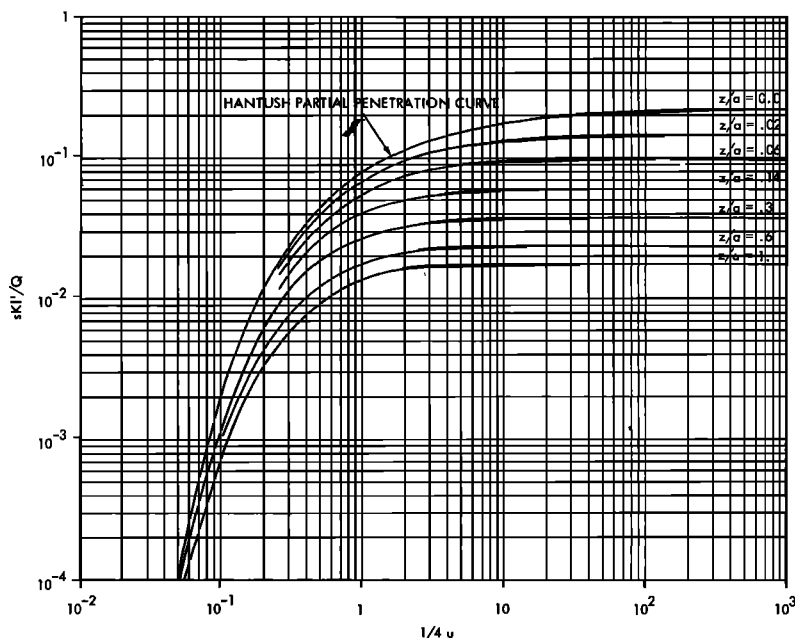


Fig. 2. Leaky partial penetration type curves tabulated from the function $W(u, z/r, r/l', z/a)$; $z/r = 0.1$, and $r/l' = 0.5$.

CONCLUSION

Solutions have been obtained for four systems of well completion. A routine to tabulate the flow systems most often encountered is available. Within the limitations of the initial assumptions, good results can be achieved by using the technique.

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